ASSIGNMENT 3

DBMS [ENCS204]

SUBMITTED BY:- YUKTI VADEHRA

SUBMITTED TO:- Ms MANSI KAJAL

ROLLNO:- 2401010128

COURSE:- B.TECH CSE CORE (SEC-A)

2401010128

UNIT-3

SECTION-A

Q1 What is SQL? Why is it important in DBMS?

Ans. SQL (Structured Query Language) is the standard language used to communicate with Relational Database Management System (RDMS).

→ It allows users to perform tasks such as creating tables, updating data, retrieving specific information with complex filters

→ Managing databases – security/permissions.

Q2 What is the difference between DDL and DML?

Ans. DDL (Data Definition Language) :- used to define or change the database, structure (schema)

eg Create, Alter, Drop.

DML (Data Manipulation Language) :- used to manage and manipulate data which is in those structures

eg. Insert, Update, Delete.

Q3. What is a Join? Explain its types.

Ans. A JOIN is used to combine rows from two or more tables based on a related column between them.

→ INNER JOIN:- Returns records that have matching values in both sides.

→ LEFT (OUTER) JOIN:- Returns all records from the left table and matched records from the right.

→ RIGHT (OUTER) JOIN:- Returns all records from the right table and matched records from the left.

→ FULL (OUTER) JOIN:- Returns all records from when there is a match in either the left or right table.

Q4. What is a GROUP BY and HAVING clause?

Ans. → GROUP BY:- Collapses multiple rows into summary rows. eg. Find the no. of students in city. It is used with aggregate functions like COUNT, MAX, SUM.

→ HAVING :- Acts as a filter for the groups created by GROUP BY. You can't use WHERE with aggregate functions, instead HAVING is used.

Q5. What is a Subquery?

Ans. A subquery is a query nested inside another SQL statement (the outer query) It is usually used in the "where" or "from" clause to provide a data set that the outer query uses for further filtering or calculation.

Q6. What is relational algebra?

Ans. It is a procedural query language that provides a theoritical foundation for relational databases. It uses operators like σ for select, π for projection & joins for joins to perform operations on relations (tables) or produce new relations.

Q7. What is the difference between WHERE & HAVING?

Ans.

WHERE	HAVING
→ Filters rows ~~f~~ before any grouping is performed	→ Filters groups after the GROUPING is performed (clause has been applied).

SECTION-B

Q1 Write an SQL query to retrieve Students with marks > 75.

Ans. SELECT * FROM students WHERE marks > 75;

Q2. WAQ using JOIN to combine Student and Course tables.

Ans. SELECT Students.Name, Course.CourseName FROM Student INNER JOIN Course ON Student.CourseID = Course.CourseID;

Q3. WAQ using GROUP BY to count Students per department.

Ans SELECT Department, COUNT(*) AS Total Students FROM Students GROUP BY Department;

SECTION-C

Q1. A Company database stores employee and department data. WAQ to find employees in a specific department.

Ans. SELECT E.EmpName, D.DeptName FROM Employee E JOIN Departments D on E.DeptID = D.DeptID WHERE D.DeptName = 'Engineering';

- Join is used to Connect Employee table with Department table.
WHERE clause to filter for the department name.

Q2. E-commerce System: Retrieve total sales per customer.

Ans. SELECT CustomerID, SUM(TotalAmount) AS TotalSalesValue FROM orders GROUP BY CustomerID;

- Aggregate function (SUM) with GROUP BY is used to consolidate all individual orders belonging to a single customer ID.

Q3. A university needs top 3 students WAQ.

Ans. SELECT StudentName, GPA FROM Students ORDER BY GPA DESC LIMIT 3;

- To find top entries, data must be sorted in descending order (DESC) based on GPA and limit should be set to 3 rows.

SECTION-D

Q1. Write an SQL Query for any real life system and explain logic.

Ans. Query for "Hospital Management System":-

```
SELECT * FROM Appointments WHERE
Appointment Date = "13-04-2026";
```

LOGIC:- This allows the front desk to manage the daily schedule and resource allocation (doctors/rooms) for the upcoming day, ensuring patient flow is optimized.

Q2. Write a JOIN query and mention one use case.

Ans. Query:-

```
Select Products.ProductName, Categories.CategoryName
FROM Products INNER JOIN Categories ON
Products.CategoryID = Categories.CategoryID;
```

- Use Case:-

This is essential for retail inventory system. By linking these tables, a manager can see exactly which products belong to which category.

Q3. Write a GROUP BY Query and explain its output.

Ans. Query:- SELECT City, COUNT(StoreID)
FROM RetailStores GROUP BY City;

Explanation:- The output will provide a list of cities and the count of stores in each.